



DAFFODIL INTERNATIONAL UNIVERSITY

Department of Computing & Information System

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Submitted by

Submitted to

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Task 1 – Q1

OSI Reference Model in a Digital Banking Data Center

A Digital Banking Data Center uses the **OSI (Open Systems Interconnection) Model** to ensure secure, reliable, and efficient communication between customers, banking servers, databases, and network devices.

Layer 7 – Application Layer

This layer provides banking services such as online banking, mobile banking, ATM transactions, and internet banking. It uses secure applications that allow customers to access banking services.

Layer 6 – Presentation Layer

This layer encrypts and decrypts banking data using SSL/TLS encryption. It also converts data into a readable format to protect sensitive customer information.

Layer 5 – Session Layer

This layer creates, manages, and ends secure communication sessions between customers and the banking servers. It keeps user sessions active until the transaction is completed.

Layer 4 – Transport Layer

The Transport Layer uses the **TCP protocol** to provide reliable communication. It ensures that banking transactions are delivered accurately without losing data.

Layer 3 – Network Layer

The Network Layer uses **IP addressing and routers** to deliver data packets between branches, ATMs, customers, and the central data center.

Layer 2 – Data Link Layer

This layer transfers data between devices inside the local network using **switches** and **MAC addresses**. It also supports **VLANs** to separate departments such as Banking Servers, Database Servers, Security, and Administration.

Layer 1 – Physical Layer

The Physical Layer consists of **fiber optic cables, Ethernet cables, routers, switches, servers, and network interface cards (NICs)** that physically transmit data.

Network Type

The proposed banking network uses both:

- **LAN (Local Area Network)** inside the data center for communication among servers and network devices.
- **WAN (Wide Area Network)** to connect bank branches, ATMs, disaster recovery sites, and remote users with the central data center.

This combination provides fast internal communication and secure connectivity across different locations.

Network Architecture

The most suitable architecture is the **Client–Server Architecture**.

Justification

- Centralized management of customer data.
- Strong authentication and access control.
- Easy backup and disaster recovery.
- High security for financial transactions.
- Better scalability as more branches and users are added.

Peer-to-peer architecture is not appropriate because it lacks centralized control and security required for banking systems.

Network Topology

The recommended topology is the **Star Topology**.

Justification

- All devices connect to central switches.
- Easy to manage and troubleshoot.
- High performance.
- Failure of one cable affects only one device.
- Supports redundancy by using core and backup switches.
- Widely used in enterprise data centers.

Conclusion

The proposed Digital Banking Data Center follows all seven OSI layers to ensure secure and reliable communication. A combination of **LAN and WAN**, **Client–Server Architecture**, and **Star Topology** provides high performance, scalability, security, and fault tolerance, making it suitable for modern digital banking.

Task 1 – Q2

Network Protocols Used in a Digital Banking Data Center

A secure digital banking data center depends on several network protocols to ensure reliable, secure, and efficient communication. The following protocols operate at different OSI/TCP-IP layers.

1. HTTPS (Application Layer)

Purpose:

HTTPS (Hypertext Transfer Protocol Secure) provides secure communication between customers and the banking web server.

Role in the Banking Network:

- Encrypts online banking transactions.
- Protects usernames, passwords, and financial information.
- Prevents unauthorized users from intercepting sensitive data.
- Ensures secure access to internet and mobile banking services.

2. TCP (Transport Layer)

Purpose:

TCP (Transmission Control Protocol) ensures reliable and error-free data transmission.

Role in the Banking Network:

- Guarantees that banking transactions are delivered completely and in the correct order.
- Detects and retransmits lost packets.
- Maintains a stable connection during financial transactions.
- Provides reliable communication between clients and servers.

3. IP (Network Layer)

Purpose:

Internet Protocol (IP) provides logical addressing and routes data between different networks.

Role in the Banking Network:

- Assigns IP addresses to devices.
- Routes data between bank branches, ATMs, and the central data center.
- Enables communication across LANs and WANs.
- Supports scalable network expansion.

Conclusion

HTTPS, TCP, and IP work together to provide a secure, reliable, and efficient communication system for the Digital Banking Data Center. HTTPS protects customer data, TCP guarantees reliable transmission, and IP ensures accurate routing across the banking network.

Task 2 – Q3

IP Addressing Scheme for a Digital Banking Data Center

To build a secure and scalable banking network, the private IP block **192.168.10.0/24** is selected. Each department is assigned a separate subnet and VLAN to improve security, performance, and management.

Subnetting Table

<u>Department</u>	<u>VLAN ID</u>	<u>Network Address</u>	<u>Subnet Mask</u>	<u>Usable Host Range</u>	<u>Broadcast Address</u>	<u>Hosts</u>
Management	VLAN 10	192.168.10.144/28	255.255.255.240	192.168.10.145 – 192.168.10.158	192.168.10.159	14
Research	VLAN 20	192.168.10.64/26	255.255.255.192	192.168.10.65 – 192.168.10.126	192.168.10.127	30
Security & Monitoring	VLAN 40	192.168.10.128/28	255.255.255.240	192.168.10.129 – 192.168.10.142	192.168.10.143	14
ATM booth	VLAN 40	192.168.10.192/26	255.255.255.192	192.168.10.193 – 192.168.10.254	192.168.10.255	14
Accounting	VLAN 50	192.168.11.0/26	255.255.255.240	192.168.11.1 – 192.168.11.158	192.168.11.159	14
Finance	VLAN 60	192.168.11.64/26	255.255.255.192	192.168.11.65 – 192.168.11.126	192.168.11.127	6

Server Farm	VLAN 70	192.168.10.0/26	255.255.255.192	192.168.10.1 – 192.168.10.62	192.168.10.63	6
Database Servers	VLAN 80	192.168.10.64/27	255.255.255.224	192.168.10.65 – 192.168.10.94	192.168.10.95	30
ICT	VLAN 90	192.168.12.128/26	255.255.255.192	192.168.12.129 – 192.168.12.190	192.168.12.191	14
GST	VLAN 100	192.168.12.128/26	255.255.255.192	192.168.12.1 – 192.168.12.62	192.168.12.63	14
Administration	VLAN 110	192.168.10.96/27	255.255.255.224	192.168.10.97 – 192.168.10.126	192.168.10.127	14
Server Room	VLAN 120	192.168.12.192/26	255.255.255.192	192.168.12.193 – 192.168.12.254	192.168.12.255	30

VLAN Segmentation Plan

<u>VLAN</u>	<u>Department</u>	<u>Purpose</u>
VLAN 10	Server Farm	Hosts application and web servers.
VLAN 20	Database Servers	Stores customer accounts and transaction records securely.
VLAN 30	Administration	Used by bank employees and administrative staff.

VLAN 40	Security & Monitoring	Supports firewalls, IDS/IPS, CCTV, and monitoring systems.
VLAN 50	Management	Reserved for network administrators to manage devices securely.
VLAN 60	Backup & Disaster Recovery	Used for backup servers and disaster recovery operations.

Justification

The proposed VLAN design provides several advantages:

- Separates sensitive departments into different broadcast domains.
- Improves network security by restricting unnecessary communication.
- Reduces broadcast traffic and increases network performance.
- Simplifies network management and troubleshooting.
- Supports future expansion by allowing additional VLANs to be added easily.
- Protects critical banking systems from unauthorized access.

Conclusion

The proposed IP addressing and VLAN segmentation plan provides a secure, scalable, and well-organized network for a Digital Banking Data Center. Each department has its own subnet and VLAN, improving security, performance, and overall network efficiency.

Task 2 – Q4

Comparison of Routing Protocols

A Digital Banking Data Center connects multiple bank branches, ATMs, disaster recovery (DR) sites, and the central data center. To ensure reliable communication, two routing protocols are considered: **RIP** and **OSPF**.

Comparison Table

<u>Feature</u>	<u>RIP (Routing Information Protocol)</u>	<u>OSPF (Open Shortest Path First)</u>
Routing Type	Distance Vector	Link State
Metric	Hop Count	Cost (Bandwidth-Based)
Maximum Network Size	Small Networks	Large Enterprise Networks
Maximum Hops	15	No Practical Hop Limit
Convergence Speed	Slow	Fast
Scalability	Low	High
Resource Usage	Low	Moderate

Best Use

Small Offices

Banking Data Centers & Enterprise
Networks

Selected Routing Protocol: OSPF

OSPF is the most appropriate routing protocol for the Digital Banking Data Center.

Justification

- Provides **fast convergence**, reducing downtime during network failures.
- Supports **large enterprise networks** with multiple branches.
- Selects the **best path based on link cost and bandwidth**.
- Supports **hierarchical network design using areas**.
- Offers **high scalability** for future expansion.
- Delivers **stable and reliable routing**, which is essential for financial transactions.

For these reasons, OSPF is more suitable than RIP for a banking environment.

Key Routing Configuration Decisions

1. OSPF Area Design

- Configure the **Data Center as Area 0 (Backbone Area)**.
- Connect branch offices and disaster recovery sites through Area 0 to ensure efficient routing.

2. Cost Metric

- Allow OSPF to select the path with the **lowest cost**, which usually corresponds to higher-bandwidth links such as fiber optic connections.

3. Route Advertisement

- Advertise all internal VLAN subnets (Server Farm, Database, Administration, Security, Management, and Backup) through OSPF.

4. Route Summarization

- Summarize routes where possible to reduce routing table size and improve router performance.

5. Route Redistribution

- If a legacy network using RIP still exists, configure **controlled route redistribution** so that RIP and OSPF networks can communicate during migration.

Conclusion

Although RIP is simple to configure, it is not suitable for a large banking network because of its limited scalability and slower convergence. OSPF provides faster routing, better scalability, efficient path selection, and high reliability, making it the ideal routing protocol for a Digital Banking Data Center.

Task 3 – Q5

Application and Network Protocol Analysis

For the proposed **Digital Banking Data Center**, several application and network protocols are required to provide secure and reliable banking services. Two important protocols are **HTTPS** and **DNS**.

1. HTTPS – Hypertext Transfer Protocol Secure

HTTPS is mainly used for secure communication between customers and banking web servers.

Reliability:

HTTPS normally works over TCP, which provides reliable and ordered delivery of data. This is important for online banking transactions because data must reach the server correctly.

Performance:

HTTPS provides efficient communication for web-based banking services. Although encryption introduces some processing overhead, modern servers can handle secure connections efficiently.

Security:

Security is the main advantage of HTTPS. It uses TLS to encrypt communication and helps protect sensitive information such as login credentials, account details, and transaction data from interception.

Use Case:

HTTPS can be used for:

- Internet banking portals
- Customer login
- Online fund transfers
- Account information
- Secure communication between users and banking web servers

2. DNS – Domain Name System

DNS translates human-readable domain names into IP addresses so that users can reach the appropriate banking services.

Reliability:

DNS allows clients to locate the correct server using its domain name. Redundant DNS servers can improve availability if one DNS server becomes unavailable.

Performance:

DNS caching reduces repeated queries and helps users reach banking services faster. A properly designed DNS infrastructure can therefore reduce lookup delays.

Security:

DNS can be targeted by attacks such as DNS spoofing or cache poisoning. Therefore, the banking environment should use appropriate DNS security controls and trusted DNS infrastructure.

Use Case:

DNS can be used to:

- Resolve the banking website domain to its server IP address.
- Locate internal banking services.
- Support communication between different network services.

Comparison of HTTPS and DNS:

<u>Characteristic</u>	<u>HTTPS</u>	<u>DNS</u>
Main Function	Secure web communication	Domain-name resolution
Main Layer	Application Layer	Application Layer
Reliability	High when used with TCP	Depends on DNS server availability and design
Performance	Secure but encryption adds some processing overhead	Fast because of caching
Security Importance	Protects sensitive banking communication	Helps securely locate network services
Banking Use	Online banking and transactions	Finding banking web/internal services

Why Both Protocols Are Important

HTTPS and DNS perform different but complementary functions. **DNS helps users and systems locate the required banking services, while HTTPS protects the actual web communication between the client and the banking server.** Using both properly helps the proposed network provide reliable, efficient, and secure digital banking services.

Conclusion

HTTPS is particularly important for protecting confidential banking communication, while DNS is essential for locating banking services efficiently. Therefore, both protocols should be properly designed and secured within the Digital Banking Data Center.

Task 3 – Q6

Network Media, Connectors and Hardware Analysis

For the proposed **Digital Banking Data Center**, the physical network infrastructure must provide high speed, reliability, security, and fault tolerance. The major components selected for the design are fiber-optic cables, Cat6A Ethernet cables, switches, routers, firewalls, and wireless access points.

1. Fiber-Optic Cable

Fiber-optic cable is recommended for high-speed connections between the **core network, server farm, and other critical network zones.**

Reasons for selection:

- Supports high bandwidth.
- Suitable for long-distance connections.
- Less affected by electromagnetic interference.
- Provides reliable communication for critical banking infrastructure.

Connector: LC or similar fiber connector can be used for data-center fiber connections.

2. Cat6A Ethernet Cable

Cat6A cables can be used for connecting servers and network devices over shorter distances inside the data center.

Reasons for selection:

- Provides high-speed Ethernet connectivity.
- Easier to install than fiber for short connections.
- Suitable for server and switch connections.
- Cost-effective for internal network connections.

Connector: RJ-45 connector.

3. Core and Access Switches

Core switches form the central switching layer of the data center, while **access switches** connect servers and other endpoint devices.

The switches can support VLANs such as:

- VLAN 10 – Server Farm
- VLAN 20 – Database Servers
- VLAN 30 – Administration
- VLAN 40 – Security & Monitoring
- VLAN 50 – Management
- VLAN 60 – Backup & Disaster Recovery

This separation reduces unnecessary broadcast traffic and improves network security.

4. Routers

Routers are required to connect the data center with external networks such as **bank branches, ATM networks, and other sites.**

They perform Layer 3 routing and can run **OSPF** to dynamically select appropriate paths between different networks.

5. Firewall

A firewall should be positioned between the external network and the internal banking infrastructure.

Its major functions include:

- Filtering unauthorized traffic.
- Controlling inbound and outbound connections.
- Protecting critical servers.
- Enforcing security policies.
- Helping prevent unauthorized network access.

For a banking environment, redundant firewalls can also improve availability.

6. Wireless Access Points

Wireless Access Points can provide controlled Wi-Fi access for authorized employees and administrators.

They should **not provide unrestricted access to critical database or server VLANs**. Wireless users should be placed into appropriate VLANs and controlled using authentication and access policies.

Proposed Network Architecture

Physical Connectivity

Component

Recommended Medium

Main Purpose

Router ↔ Firewall	Fiber/Ethernet	External network connection
Firewall ↔ Core Switch	Fiber	High-speed secure backbone
Core ↔ Access Switch	Fiber	High-speed internal connectivity
Switch ↔ Server	Cat6A/Fiber	Server connectivity
Access Point ↔ Switch	Cat6A	Wireless connectivity
Core ↔ Backup/DR	Fiber/WAN link	Disaster recovery connectivity

Design Justification

The combination of **fiber-optic backbone links, Cat6A Ethernet, Layer 3 switches, routers, firewalls, and controlled wireless access points** creates a reliable physical infrastructure for the banking data center.

Fiber is used for high-speed critical links, while Cat6A is suitable for shorter internal connections. VLAN-capable switches provide logical segmentation, routers provide inter-network connectivity, and firewalls enforce security boundaries. This design therefore supports the assignment's requirements for a reliable and efficient physical implementation.

Task 4 – Q7 (4 Marks)

Wireless Network Evaluation for a Digital Banking Data Center

For the proposed Digital Banking Data Center, wireless connectivity should be designed mainly for **authorized employees, administrators, and controlled operational devices**. Critical database and transaction servers should remain on wired, isolated network segments.

1. Wireless Standard

Wi-Fi 6 (IEEE 802.11ax) is selected for the organization's wireless network.

It is suitable because it provides:

- Better performance in high-density environments.
- More efficient use of the wireless spectrum.
- Improved capacity when many users are connected.
- Better support for modern laptops, tablets, and mobile devices.
- More efficient communication between access points and multiple clients.

2. Wireless Devices

The main wireless devices required are:

- **Wi-Fi 6 Access Points (APs)** – provide wireless connectivity.
- **Wireless LAN Controller (WLC)** – centrally manages multiple APs.
- **PoE-enabled switches** – provide both network connectivity and power to APs.
- **Authentication server** – controls access for authorized users.

3. Access Point Placement

Access points should not simply be installed at equal distances. Their locations should be determined according to:

- Building structure and wall materials.
- Number of expected users.
- Required coverage area.
- Areas with high user density.
- Sources of wireless interference.
- Required signal strength.

For example, APs can be placed in **staff offices, meeting areas, security monitoring rooms, and other authorized operational areas**.

Critical server rooms should prioritize wired connectivity rather than depending on Wi-Fi.

4. Coverage and Capacity

Both **coverage and capacity** must be considered.

A single AP should not be overloaded with too many users. High-density areas should receive additional AP capacity rather than simply increasing transmission power.

The network administrator should monitor:

- Number of connected users.
- Signal strength.
- Channel utilization.
- Interference.
- Data traffic.
- AP performance.

This helps maintain stable wireless service as the organization grows.

5. Channel Planning

A proper channel plan should be implemented to reduce interference between nearby APs.

- Use appropriate non-overlapping channels where applicable.
- Avoid unnecessarily placing nearby APs on the same channel.
- Monitor channel utilization regularly.
- Adjust channels when interference increases.
- Use automatic channel management where appropriate.

6. Wireless Security

Wireless access should be separated from critical banking systems using **VLANs and access-control policies**.

For example:

<u>Wireless Network</u>	<u>Purpose</u>	<u>Access</u>
Staff Wi-Fi	Employee devices	Controlled internal resources

Admin Wi-Fi	Network administrators	Restricted management resources
Guest Wi-Fi	Visitors	Internet only
IoT/Operational Wi-Fi	Approved devices	Limited authorized services

Guest users should never receive direct access to sensitive banking databases or server VLANs.

Conclusion

Wi-Fi 6 with properly positioned access points, capacity planning, channel management, and VLAN-based segmentation provides a suitable wireless solution for the Digital Banking Data Center. However, critical banking servers and databases should continue to use secure wired connections. This approach balances **performance, coverage, scalability, and security**.

Task 4 – Q8

Security Risks and Protection Mechanisms

A Digital Banking Data Center handles highly sensitive information such as customer accounts, authentication credentials, and financial transactions. Therefore, the network must be protected against unauthorized access, data interception, denial-of-service attacks, and other network threats.

1. Unauthorized Access

Unauthorized users may attempt to access internal banking servers or management systems.

Risk:

An attacker who gains access could view or modify sensitive banking information.

Protection:

Implement **strong access control and authentication**. Users should only receive the minimum network and system privileges required for their jobs. VLAN segmentation should also prevent ordinary users from directly accessing critical database and management networks.

2. Data Interception / Man-in-the-Middle Attack

Attackers may attempt to intercept communication between users and banking servers.

Risk:

Sensitive information such as login credentials or transaction data could be exposed.

Protection:

Use **HTTPS/TLS** for web-based banking communication and **VPN/IPsec** for secure connections between trusted remote sites where appropriate. Encryption helps protect data while it is travelling across the network.

3. Denial-of-Service (DoS) Attack

An attacker may send a large amount of traffic toward banking services to make them unavailable.

Risk:

Customers may be unable to access online banking or complete transactions.

Protection:

Use **firewall rules and traffic filtering** to control suspicious or unauthorized traffic. Monitoring systems can also help detect unusual traffic patterns and respond to attacks.

4. Internal Network Threats

A compromised employee device could be used to reach sensitive systems.

Risk:

An attacker may move from an employee network toward database or server networks.

Protection:

Use **network segmentation through VLANs and access-control rules**. Critical database and server VLANs should only be accessible to authorized services and users.

Recommended Security Mechanisms:

<u>Security Mechanism</u>	<u>Purpose</u>
Firewall	Filters unauthorized inbound and outbound traffic.
VPN/IPsec	Protects communication between remote sites over untrusted networks.
Access Control	Restricts users and devices according to their authorized roles.
VLAN Segmentation	Separates critical banking systems into isolated network segments.
HTTPS/TLS	Encrypts sensitive web-based banking communication.

Overall Security Strategy

The proposed network should use **multiple layers of protection rather than relying on a single security mechanism**. The firewall provides a security boundary, VLANs isolate critical systems, access control limits who can reach them, and encryption protects sensitive data during transmission.

Conclusion

A combination of **firewalls, VPN/IPsec, access control, VLAN segmentation, and encrypted communication** can significantly reduce the major security risks facing the Digital Banking Data Center. This layered approach improves confidentiality, integrity, and availability of banking services.

Task 4 – Q9

Overall Conclusion

The proposed **Secure Network Architecture for a Digital Banking Data Center** provides a balanced solution for the organization's communication, security, and future growth requirements.

The design is **scalable** because the IP addressing scheme, VLAN segmentation, and OSPF routing architecture can support additional users, servers, branches, and network services without requiring a complete redesign.

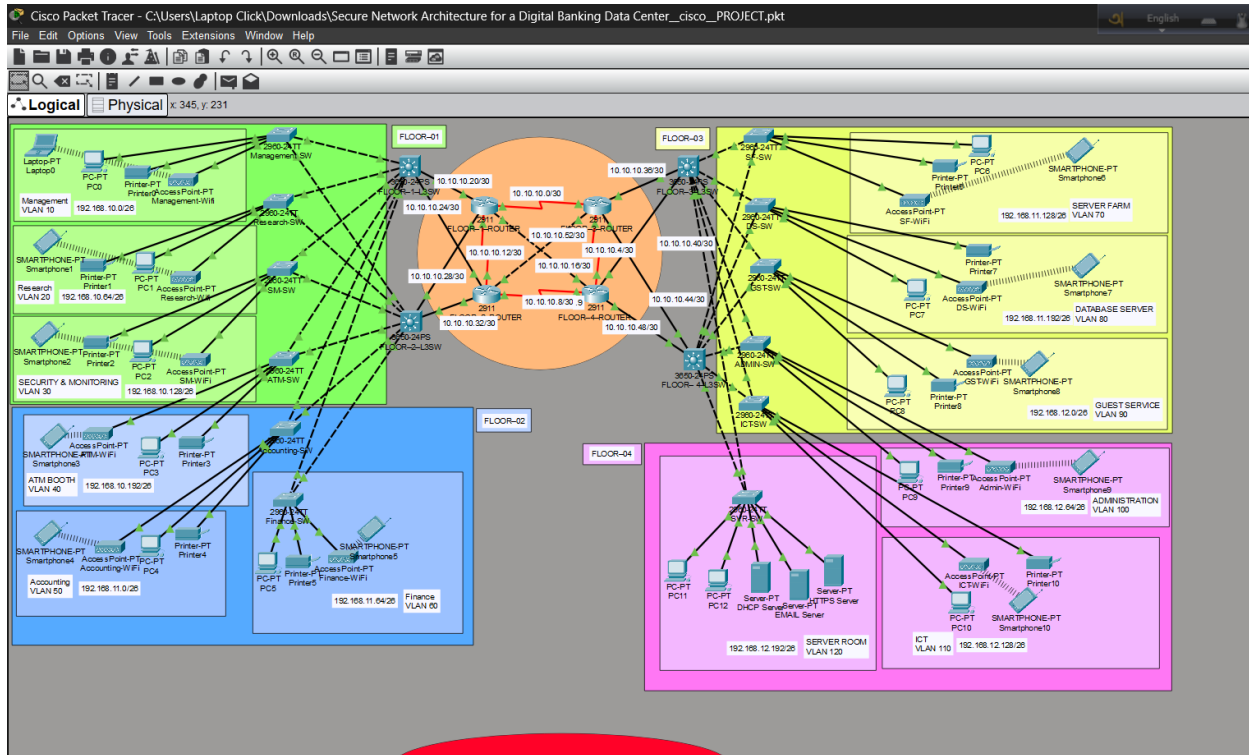
It is **secure** because critical banking resources are separated using VLANs, while firewalls, access-control mechanisms, HTTPS/TLS, and VPN/IPsec provide multiple layers of protection against unauthorized access and data interception.

The design is **reliable** because it uses high-speed network media, appropriate routing, controlled wireless access, and a structured data-center architecture. OSPF can also provide efficient path selection and faster adaptation to network changes.

Final Statement:

Overall, the proposed network provides a **scalable, secure, and reliable infrastructure** for digital banking operations. It supports current banking requirements while allowing the organization to expand its services and network infrastructure in the future without compromising security or performance.

CISCO PACKET TRACER :



TESTING and VERIFICATION :

We need to test the network to make sure everything is working correctly. 15.1 Ping Testing We can use the ping command to see if devices can talk to each other. For example we can use the command: ping [Destination IP]. If we get a response it means the devices can talk to each other.

Figure 2
connectivity test between network devices

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Smart...	Printer1	ICMP		0.000	N	5	(edit)	(delete)
	Successful	Smart...	Printer2	ICMP		0.000	N	6	(edit)	(delete)
	Successful	PC5	Smartpho...	ICMP		0.000	N	7	(edit)	(delete)